ASSIGNMENT-02

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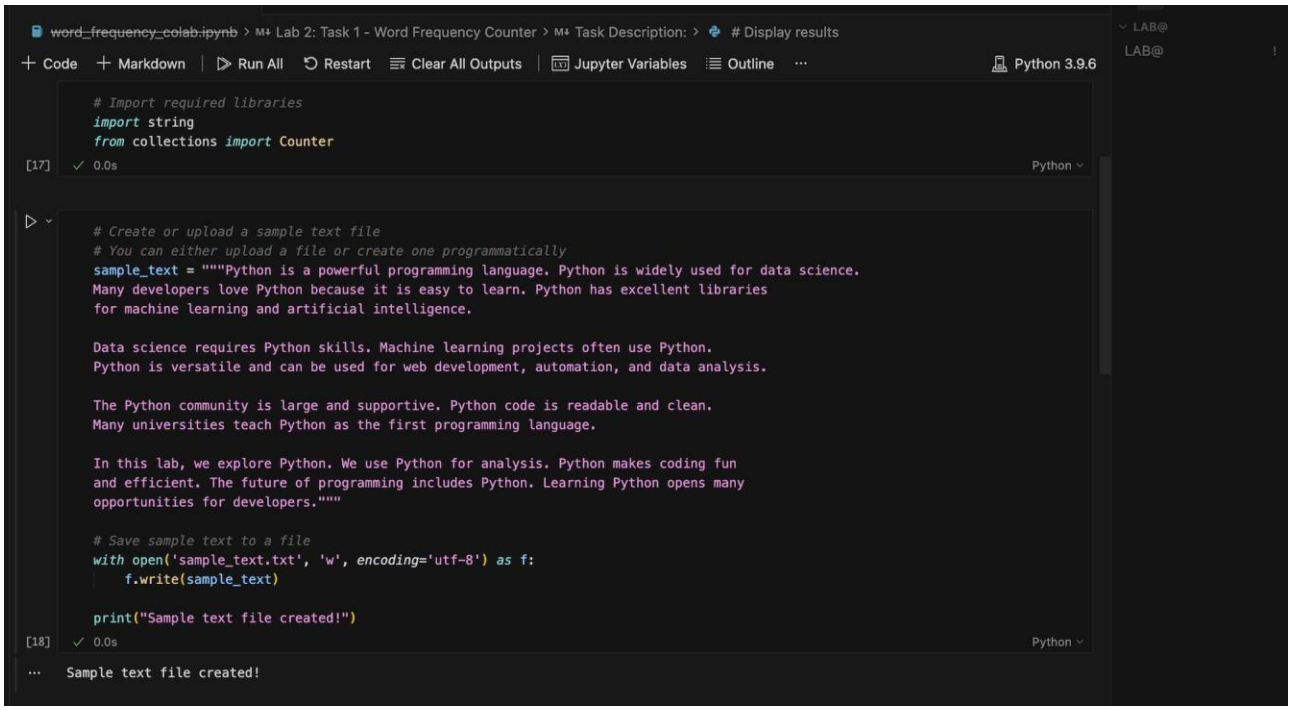
Batch:05

Q) Task 1: Word Frequency from Text File
❖ Scenario:
You are analyzing log files for keyword frequency.
❖ Task:
Use Gemini to generate Python code that reads a text file and counts word frequency, then explains the code.
❖ Expected Output:
➢ Working code
➢ Explanation
➢ Screenshot

Solution:

PROMPT
Generate a Python program in Google Colab that reads a text file and counts the frequency of each word.

CODE:



~~word_frequency_colab.ipynb~~ > Lab 2: Task 1 - Word Frequency Counter > Task Description: > # Display results

+ Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ··· Python 3.9.6

```python
def count_word_frequency(filename):
    """
    Read a text file and count the frequency of each word.

    Args:
        filename (str): Path to the text file to analyze

    Returns:
        Counter: Counter object with words as keys and frequencies as values
    """
    try:
        # Open and read the file
        with open(filename, 'r', encoding='utf-8') as file:
            text = file.read()

        # Convert to lowercase and remove punctuation
        translator = str.maketrans('', '', string.punctuation)
        text = text.translate(translator).lower()

        # Split text into words
        words = text.split()

        # Count word frequencies using Counter
        word_freq = Counter[str](words)

        return word_freq

    except FileNotFoundError:
        print(f"Error: File '{filename}' not found.")
        return None
    except Exception as e:
        print(f"Error reading file: {e}")
        return None
```

[19] ✓ 0.0s Python

```python
# Execute the word frequency analysis
filename = 'sample_text.txt'
word_freq = count_word_frequency(filename)
```

[20] ✓ 0.0s Python

```python
# Display results
if word_freq:
    print("\n" + "="*50)
    print("WORD FREQUENCY ANALYSIS")
    print("="*50)

    # Display top 20 most common words
    print("\nTop 20 Most Frequent Words:")
    print("-"*50)
    print(f"{'Word':<20} {'Frequency':<15} {'Percentage':<15}")
    print("-"*50)

    total_words = sum(word_freq.values())

    for word, count in word_freq.most_common(20):
        percentage = (count / total_words) * 100
        print(f"{word:<20} {count:<15} {percentage:.2f}%")

    print("-"*50)
    print(f"\nTotal unique words: {len(word_freq)}")
    print(f"Total words: {total_words}")
    print("="*50)
```

OUTPUT:

```
...
==================================================
WORD FREQUENCY ANALYSIS
==================================================

Top 20 Most Frequent Words:
--------------------------------------------------
Word                 Frequency       Percentage
--------------------------------------------------
python               15              13.64%
is                   6               5.45%
and                  6               5.45%
for                  5               4.55%
programming          3               2.73%
data                 3               2.73%
many                 3               2.73%
learning             3               2.73%
the                  3               2.73%
language             2               1.82%
used                 2               1.82%
science              2               1.82%
developers           2               1.82%
machine              2               1.82%
use                  2               1.82%
analysis             2               1.82%
...

Total unique words: 64
Total words: 110
==================================================
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

CODE Explanation:

This Python program works by first importing the required modules to handle punctuation removal and word counting. The text file is opened in read mode and its content is read completely. Then, all punctuation marks are removed and the text is converted to lowercase so that words are counted correctly without case differences. After that, the text is split into individual words. The Counter function is used to count the number of times each word appears in the file. The program also includes error handling to display a message if the file is not found or if any other error occurs. Finally, the word frequencies are displayed in an organized format, making the output easy to understand

Q) Task 2: File Operations Using Cursor AI

❖ Scenario:

You are automating basic file operations.

❖ Task:

Use Cursor AI to generate a program that:

➢ Creates a text file

➢ Writes sample text

➢ Reads and displays the content

❖ Expected Output:

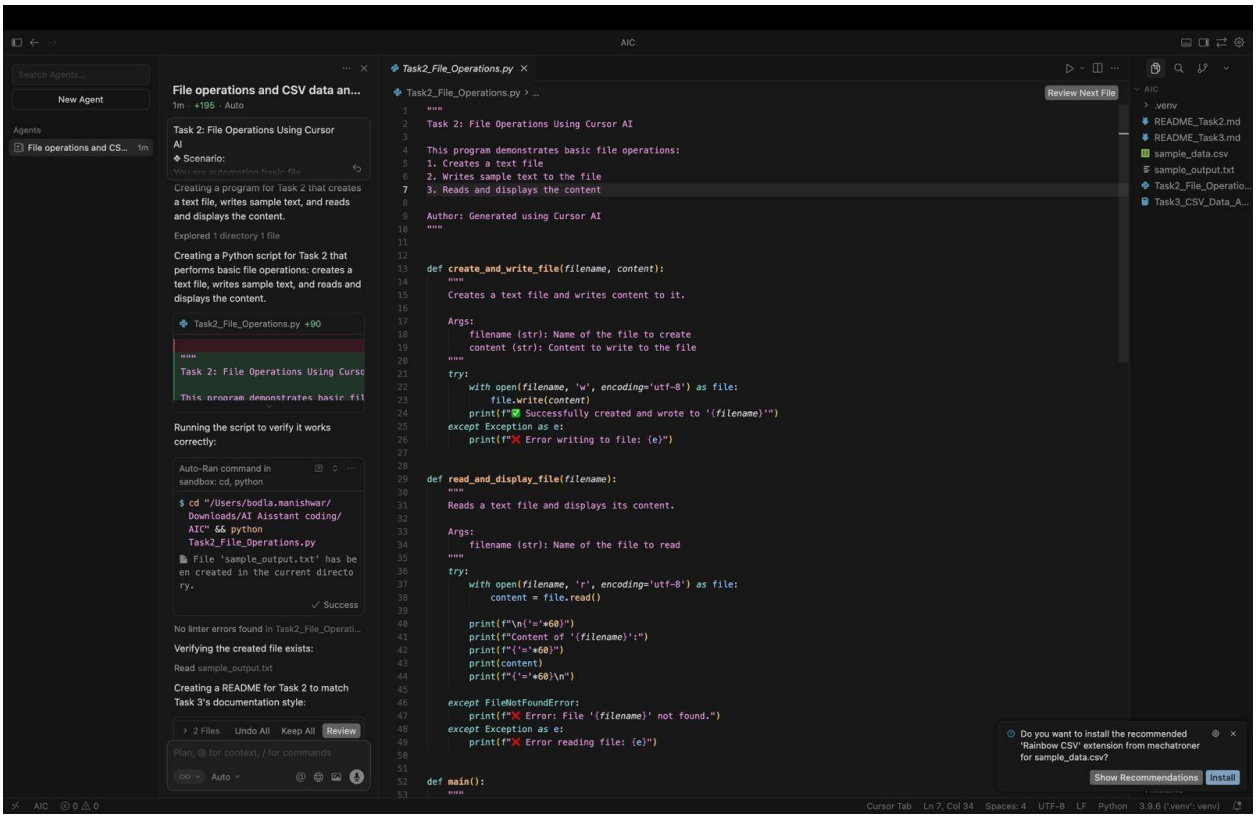
➢ Functional code

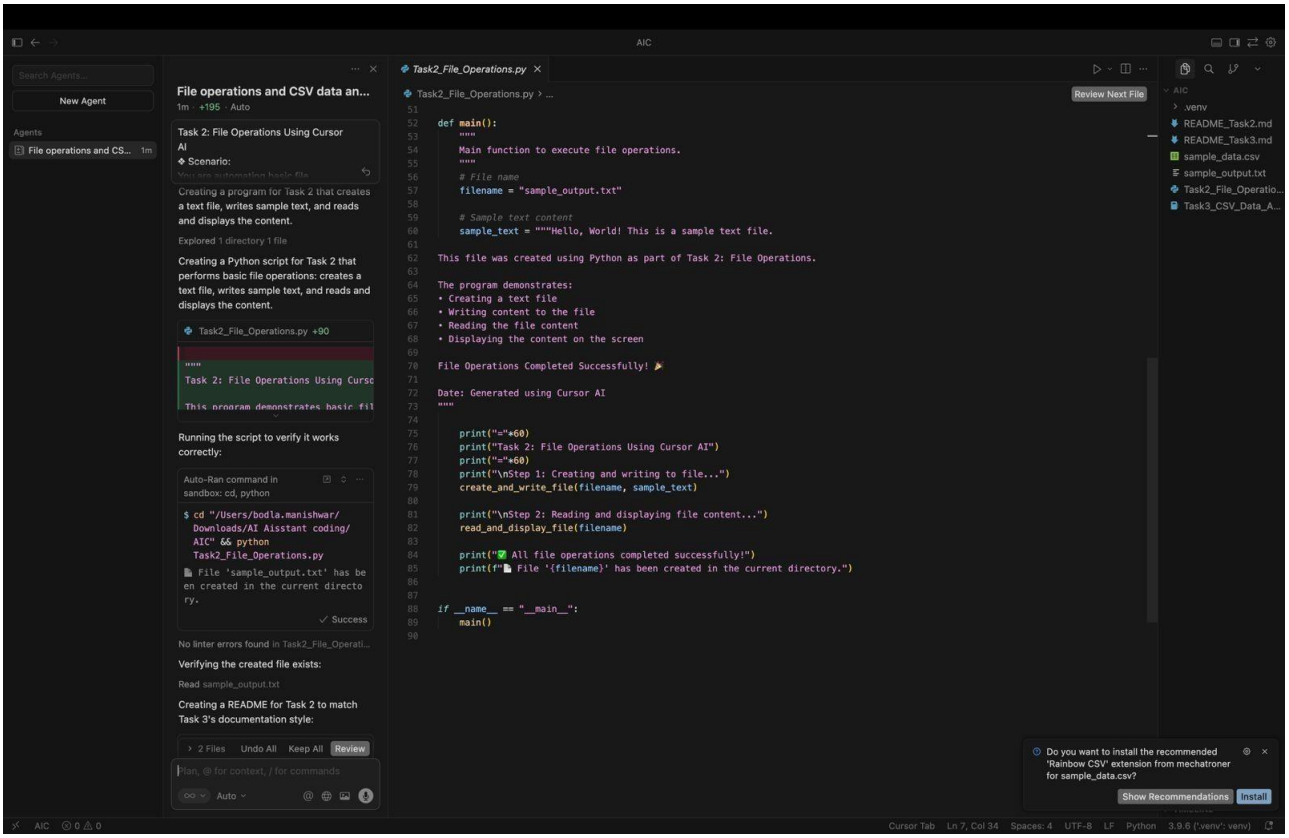
➢ Cursor AI screenshots

PROMPT:

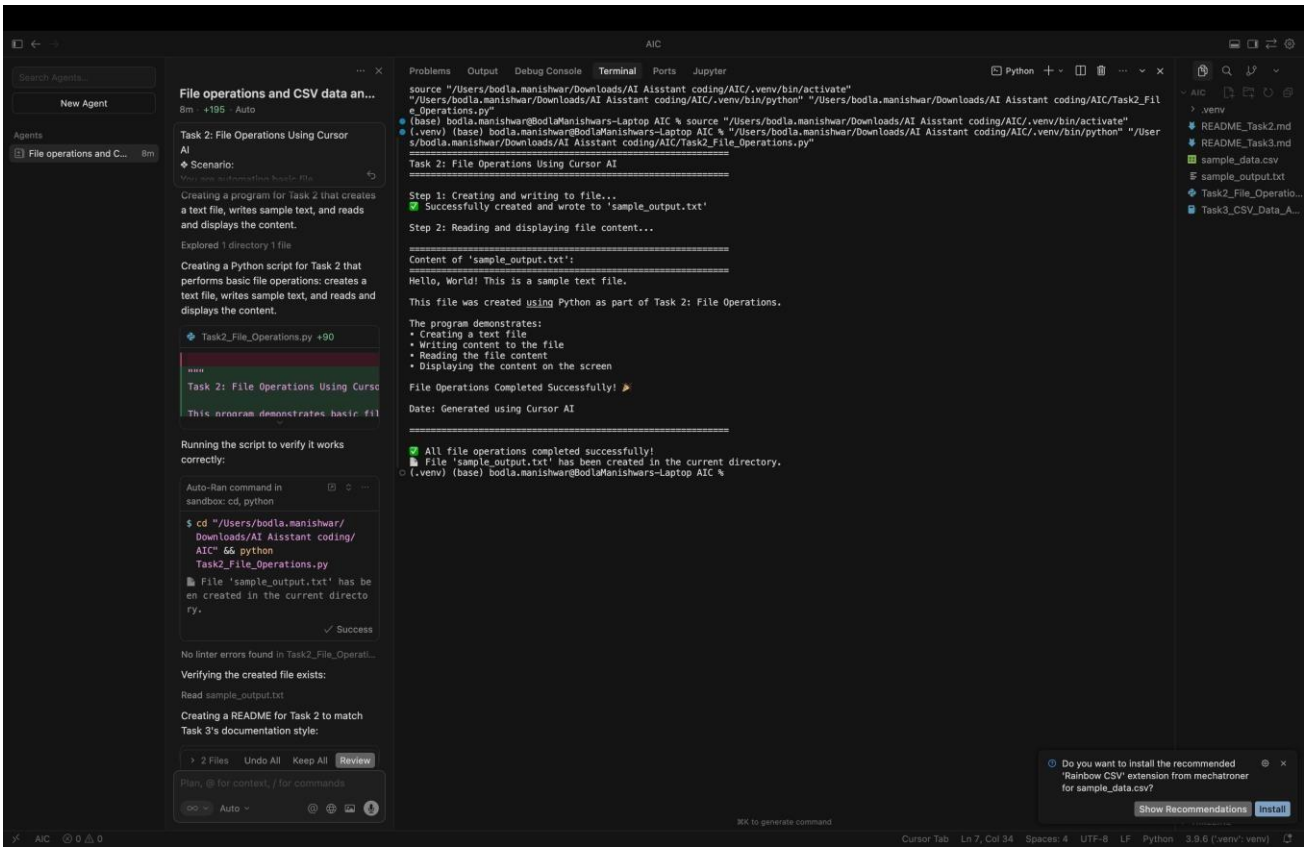
Generate a simple Python program that demonstrates basic file operations. The program should create a text file, write some sample text into it, then read the content from the file and display it on the screen.

CODE:





OUTPUT:



CODE EXPLANATION:

This Python program demonstrates basic file operations by creating a text file, writing sample content to it, and then reading and displaying that content on the screen. It uses separate functions for writing and reading files to keep the code organized and clear. The program also includes exception handling to manage errors such as file access issues, ensuring smooth execution. The main() function controls the overall flow, and the program runs only when executed directly, making it a simple and effective example of file handling in Python.

Q)Task 3: CSV Data Analysis

❖ Scenario:

You are processing structured data from a CSV file.

❖ Task:

Use Gemini in Colab to read a CSV file and calculate mean, min, and max.

❖ Expected Output:

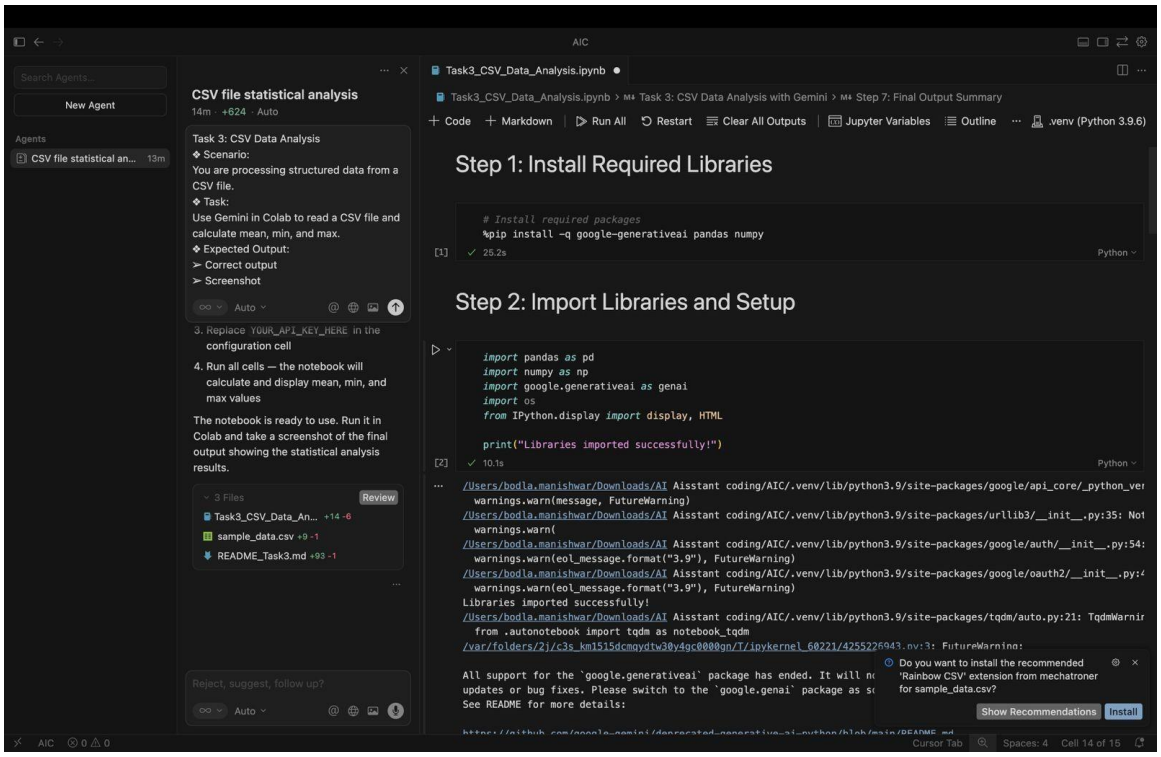
➢ Correct output

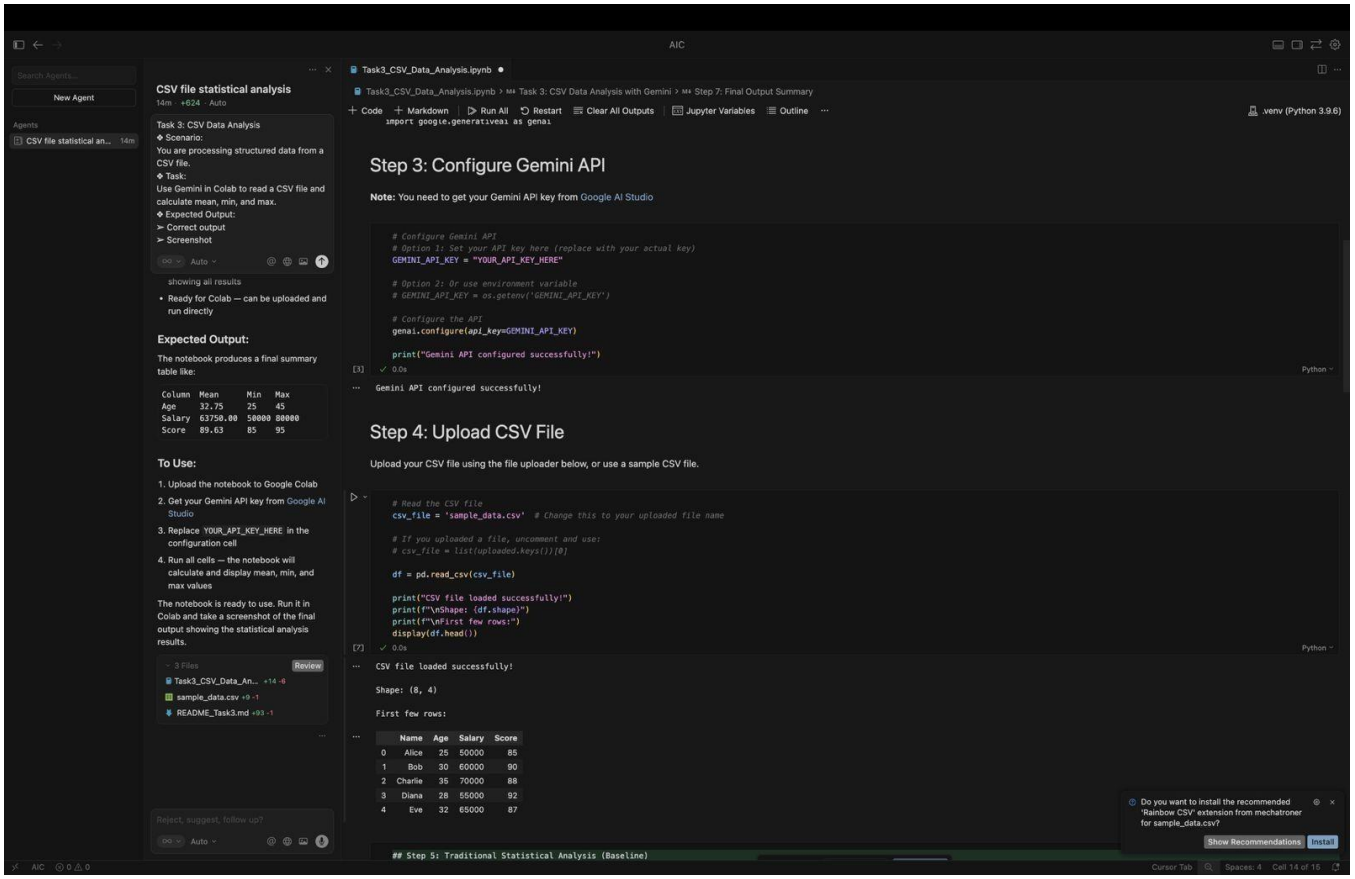
➢ Screenshot

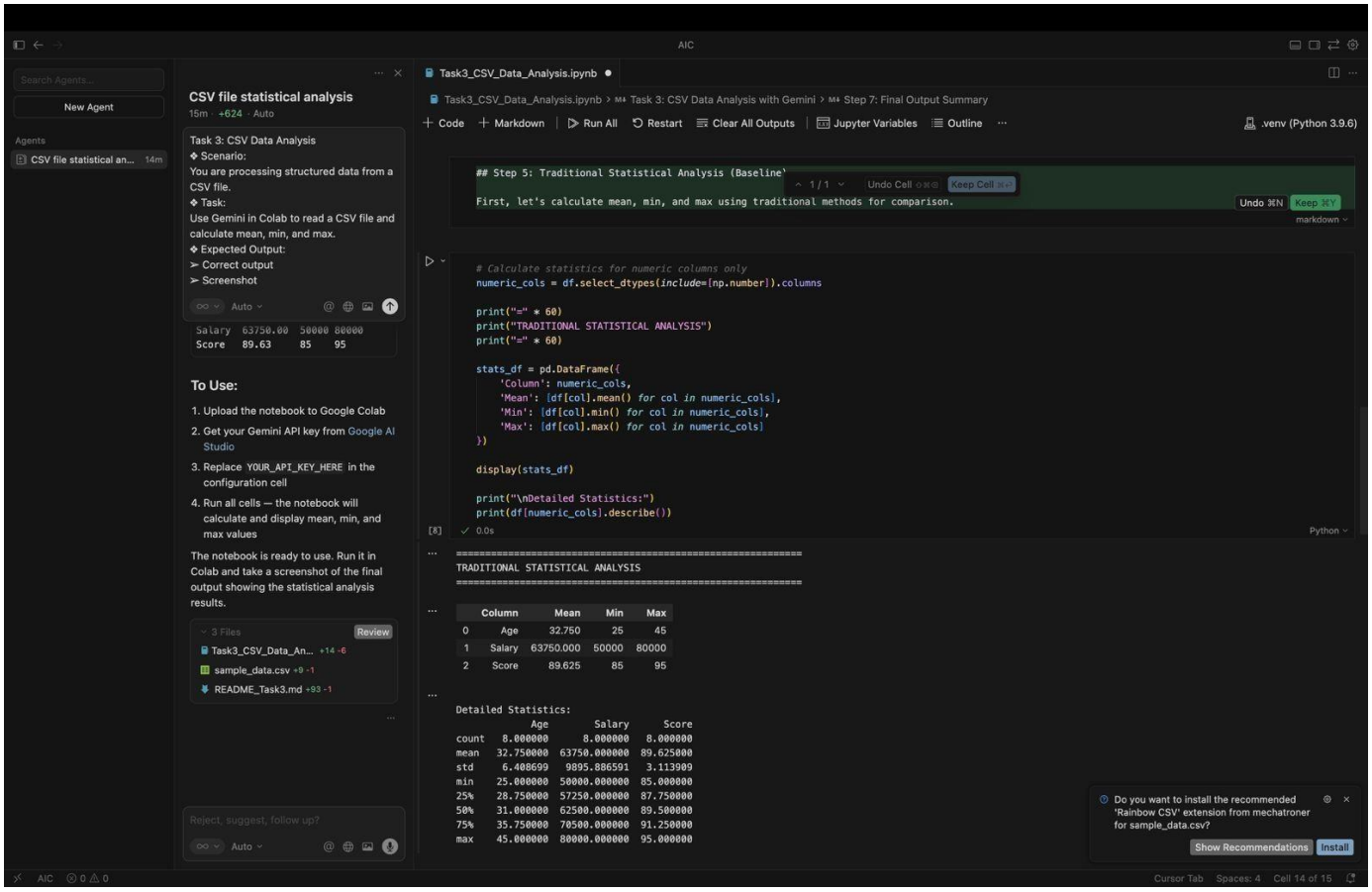
PROMPT:

Write Python code in Google Colab to read a CSV file and calculate mean, minimum, and maximum values using pandas.

CODE:




AIC
New Agent
Agents
CSV file statistical an... 14m
CSV file statistical analysis
14m · +624 · Auto
Task 3: CSV Data Analysis
❖ Scenario:
You are processing structured data from a CSV file.
❖ Task:
Use Gemini in Colab to read a CSV file and calculate mean, min, and max.
❖ Expected Output:
➢ Correct output
➢ Screenshot
showing all results
• Ready for Colab — can be uploaded and run directly
Expected Output:
The notebook produces a final summary table like:
Column Mean Min Max
Age 32.75 25 45
Salary 63750.00 50000 80000
Score 89.63 85 95
To Use:
1. Upload the notebook to Google Colab
2. Get your Gemini API key from Google AI Studio
3. Replace YOUR_API_KEY_HERE in the configuration cell
4. Run all cells — the notebook will calculate and display mean, min, and max values
The notebook is ready to use. Run it in Colab and take a screenshot of the final output showing the statistical analysis results.
3 Files
Review
Task3_CSV_Data_An... +14 -6
sample_data.csv +9 -1
README_Task3.md +93 -1
Reject, suggest, follow up?
Auto
Task3_CSV_Data_Analysis.ipynb
Task3_CSV_Data_Analysis.ipynb > Task 3: CSV Data Analysis with Gemini > Step 7: Final Output Summary
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
Outline
.venv (Python 3.9.6)
import google.generativeai as genai
Step 3: Configure Gemini API
Note: You need to get your Gemini API key from Google AI Studio
# Configure Gemini API
# Option 1: Set your API key here (replace with your actual key)
GEMINI_API_KEY = "YOUR_API_KEY_HERE"
# Option 2: Or use environment variable
# GEMINI_API_KEY = os.getenv('GEMINI_API_KEY')
# Configure the API
genai.configure(api_key=GEMINI_API_KEY)
print("Gemini API configured successfully!")
[3] 0.0s
Python
Gemini API configured successfully!
Step 4: Upload CSV File
Upload your CSV file using the file uploader below, or use a sample CSV file.
# Read the CSV file
csv_file = 'sample_data.csv' # Change this to your uploaded file name
# If you uploaded a file, uncomment and use:
# csv_file = list(uploaded.keys())[0]
df = pd.read_csv(csv_file)
print("CSV file loaded successfully!")
print(f"\nShape: {df.shape}")
print(f"\nFirst few rows:")
display(df.head())
[7] 0.0s
Python
CSV file loaded successfully!
Shape: (8, 4)
First few rows:
Name Age Salary Score
0 Alice 25 50000 85
1 Bob 30 60000 90
2 Charlie 35 70000 88
3 Diana 28 55000 92
4 Eve 32 65000 87
Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for sample_data.csv?
Show Recommendations
Install
Step 5: Traditional Statistical Analysis (Baseline)
Cursor Tab
Spaces: 4
Cell 14 of 15


AIC
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Agents
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CSV file statistical analysis
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Task 3: CSV Data Analysis
❖ Scenario:
You are processing structured data from a CSV file.
❖ Task:
Use Gemini in Colab to read a CSV file and calculate mean, min, and max.
❖ Expected Output:
➢ Correct output
➢ Screenshot
Salary 63750.00 50000 80000
Score 89.63 85 95
To Use:
1. Upload the notebook to Google Colab
2. Get your Gemini API key from Google AI Studio
3. Replace YOUR_API_KEY_HERE in the configuration cell
4. Run all cells — the notebook will calculate and display mean, min, and max values
The notebook is ready to use. Run it in Colab and take a screenshot of the final output showing the statistical analysis results.
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Task3_CSV_Data_Analysis.ipynb
Task3_CSV_Data_Analysis.ipynb > Task 3: CSV Data Analysis with Gemini > Step 7: Final Output Summary
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Markdown
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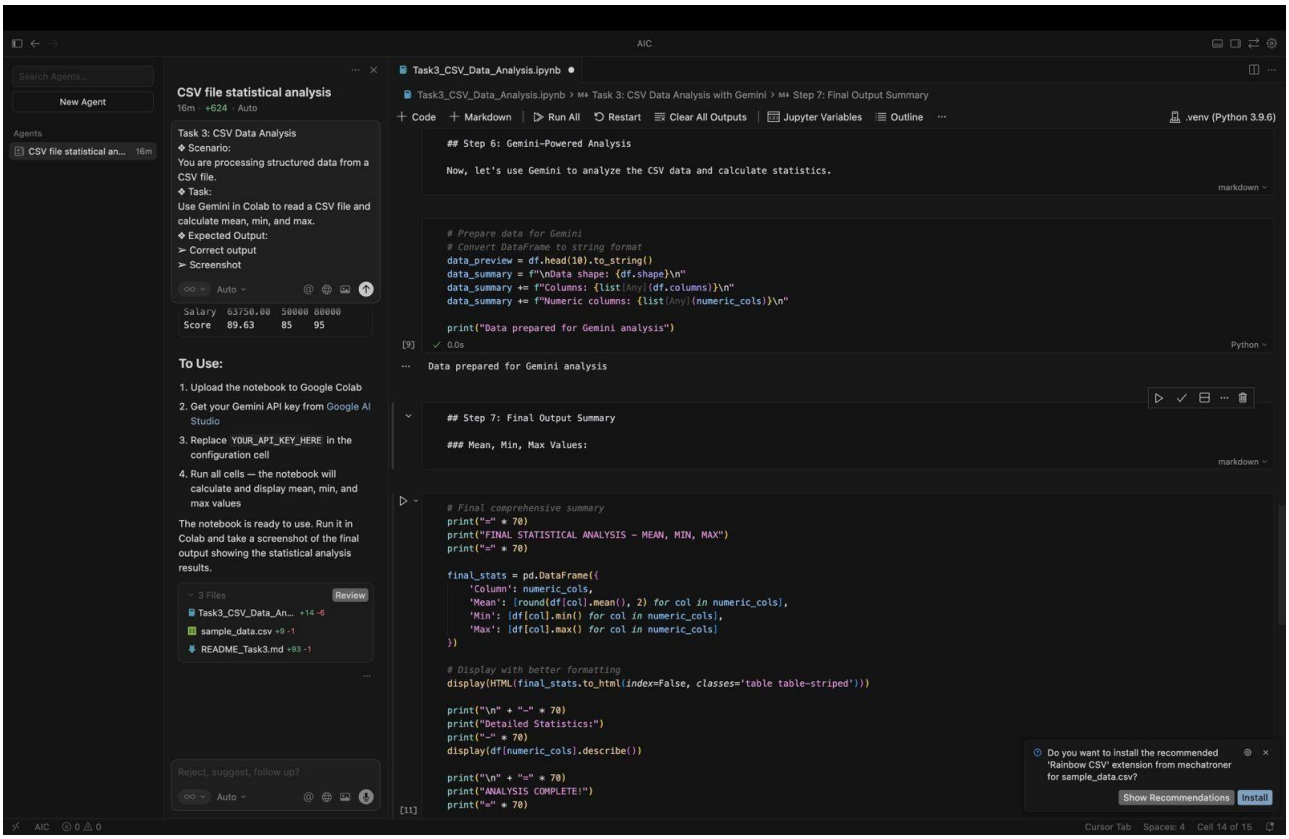
Clear All Outputs
Jupyter Variables
Outline
.venv (Python 3.9.6)
Step 5: Traditional Statistical Analysis (Baseline)
First, let's calculate mean, min, and max using traditional methods for comparison.
Undo ⌘N
Keep ⌘Y
markdown
# Calculate statistics for numeric columns only
numeric_cols = df.select_dtypes(include=[np.number]).columns
print("=" * 60)
print("TRADITIONAL STATISTICAL ANALYSIS")
print("=" * 60)
stats_df = pd.DataFrame({
'Column': numeric_cols,
'Mean': [df[col].mean() for col in numeric_cols],
'Min': [df[col].min() for col in numeric_cols],
'Max': [df[col].max() for col in numeric_cols]
})
display(stats_df)
print("\nDetailed Statistics:")
print(df[numeric_cols].describe())
[8] 0.0s
Python
TRADITIONAL STATISTICAL ANALYSIS
Column Mean Min Max
0 Age 32.750 25 45
1 Salary 63750.000 50000 80000
2 Score 89.625 85 95
Detailed Statistics:
Age Salary Score
count 8.000000 8.000000 8.000000
mean 32.750000 63750.000000 89.625000
std 6.408699 9895.886591 3.113909
min 25.000000 50000.000000 85.000000
25% 28.750000 57250.000000 87.750000
50% 31.000000 62500.000000 89.500000
75% 35.750000 70500.000000 91.250000
max 45.000000 80000.000000 95.000000
Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for sample_data.csv?
Show Recommendations
Install
Cursor Tab
Spaces: 4
Cell 14 of 15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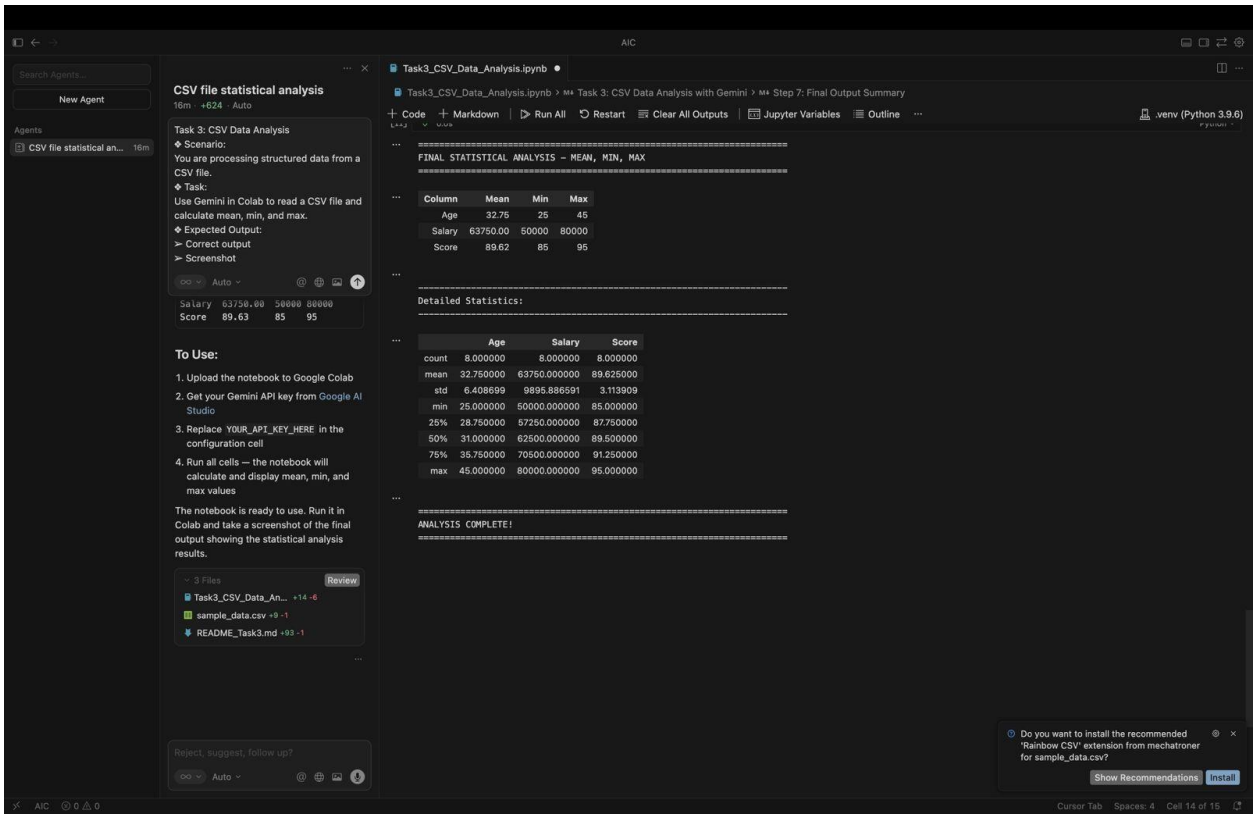
Step 6: Gemini-Powered Analysis
Now, let's use Gemini to analyze the CSV data and calculate statistics.
markdown
# Prepare data for Gemini
# Convert DataFrame to string format
data_preview = df.head(10).to_string()
data_summary = f"\nData shape: {df.shape}\n"
data_summary += f"Columns: {list(df.columns)}\n"
data_summary += f"Numeric columns: {list(numeric_cols)}\n"
print("Data prepared for Gemini analysis")
[9]
0.0s
Python
Data prepared for Gemini analysis
Step 7: Final Output Summary
Mean, Min, Max Values:
markdown
# Final comprehensive summary
print("=" * 70)
print("FINAL STATISTICAL ANALYSIS - MEAN, MIN, MAX")
print("=" * 70)
final_stats = pd.DataFrame({
'Column': numeric_cols,
'Mean': [round(df[col].mean(), 2) for col in numeric_cols],
'Min': [df[col].min() for col in numeric_cols],
'Max': [df[col].max() for col in numeric_cols]
})
# Display with better formatting
display(HTML(final_stats.to_html(index=False, classes='table table-striped')))
print("\n" + "-" * 70)
print("Detailed Statistics:")
print("-" * 70)
display(df[numeric_cols].describe())
print("\n" + "=" * 70)
print("ANALYSIS COMPLETE!")
print("=" * 70)
[11]
Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for sample_data.csv?
Show Recommendations
Install
Cursor Tab
Spaces: 4
Cell 14 of 15

OUTPUT:


AIC
Search Agents...
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CSV file statistical an... 16m
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16m · +624 · Auto
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count 8.000000 8.000000 8.000000
mean 32.750000 63750.000000 89.625000
std 6.408699 9895.886591 3.113909
min 25.000000 50000.000000 85.000000
25% 28.750000 57250.000000 87.750000
50% 31.000000 62500.000000 89.500000
75% 35.750000 70500.000000 91.250000
max 45.000000 80000.000000 95.000000
ANALYSIS COMPLETE!
Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for sample_data.csv?
Show Recommendations
Install
Cursor Tab
Spaces: 4
Cell 14 of 15

CODE EXPLANATION:

This code performs statistical analysis on numeric columns of a DataFrame (df). First, it identifies all columns that contain numerical data using select_dtypes(include=[np.number]). Then, for each numeric column, it calculates the mean, minimum, and maximum values and stores them in a new DataFrame called stats_df. This DataFrame is displayed to show a clean summary of basic statistics.

Q)Task 4: Sorting Lists – Manual vs Built-in

❖ Scenario:

You are reviewing algorithm choices for efficiency.

❖ Task:

Use Gemini to generate:

- **Bubble sort**
- **Python's built-in sort()**
- **Compare both implementations.**

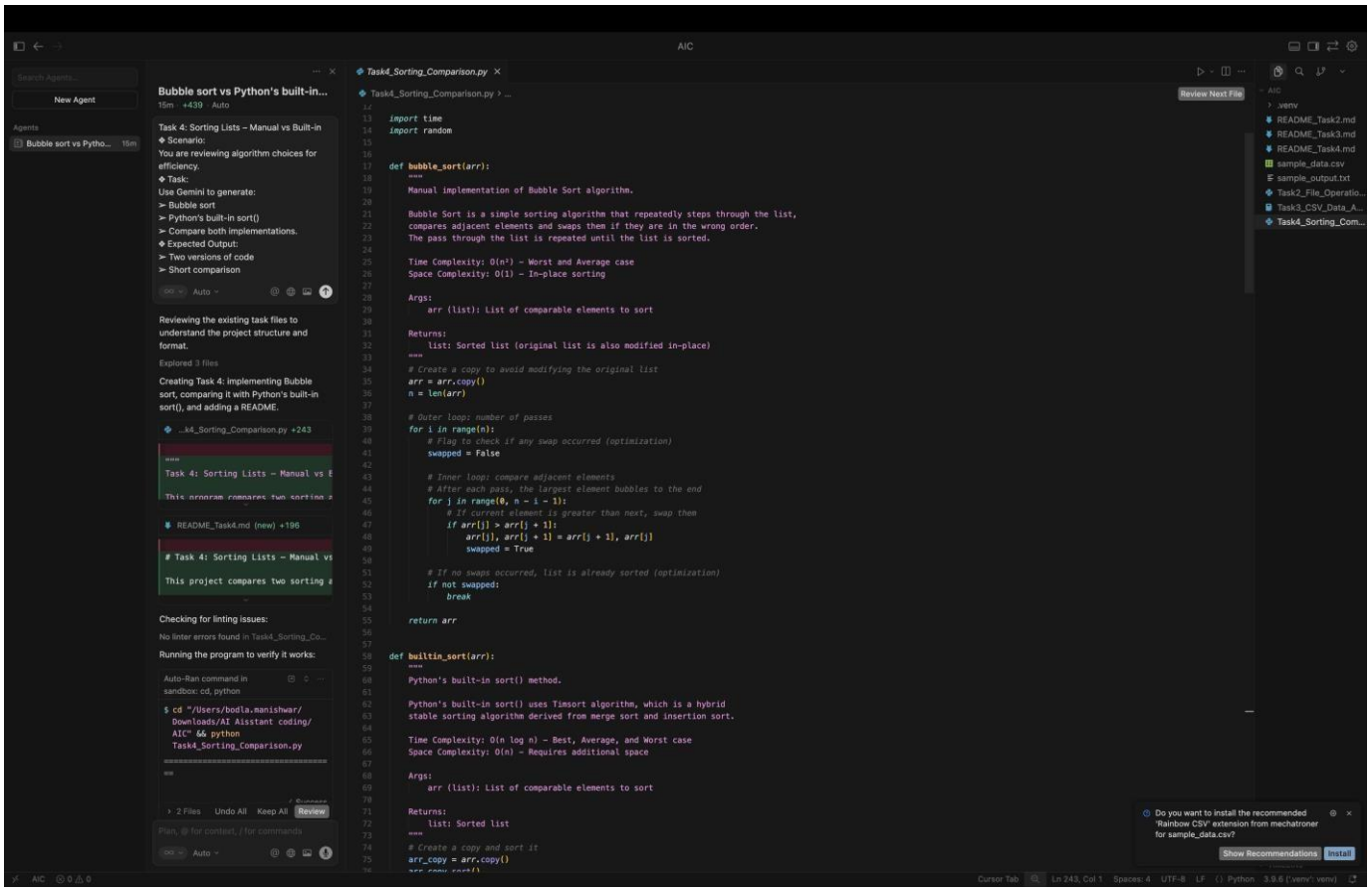
❖ Expected Output:

- **Two versions of code**
- **Short comparison**

PROMPT:

Generate Python code to sort a list using bubble sort and Python's built-in sort() method. Show both implementations clearly and provide a short comparison explaining their efficiency and usage.

CODE:



AIC

New Agent

Agents

Bubble sort vs Pytho... 16m

Bubble sort vs Python's built-in...

15m · +439 · Auto

Task 4: Sorting Lists – Manual vs Built-in
◆ Scenario:
You are reviewing algorithm choices for efficiency.
◆ Task:
Use Gemini to generate:
➢ Bubble sort
➢ Python's built-in sort()
➢ Compare both implementations.
◆ Expected Output:
➢ Two versions of code
➢ Short comparison

Auto

Reviewing the existing task files to understand the project structure and format.

Explored 3 files

Creating Task 4: implementing Bubble sort, comparing it with Python's built-in sort(), and adding a README.

...k4_Sorting_Comparison.py +243

```
"""
Task 4: Sorting Lists – Manual vs E
This program compares two sorting a
```

README_Task4.md (new) +196

```
# Task 4: Sorting Lists – Manual vs
This project compares two sorting a
```

Checking for linting issues:

No linter errors found in Task4_Sorting_Co...

Running the program to verify it works:

Auto-Ran command in sandbox: cd, python

```
$ cd "/Users/bodla.manishwar/
  Downloads/AI Aisstant coding/
  AIC" && python
  Task4_Sorting_Comparison.py
==================================
==
```

2 Files Undo All Keep All Review

Plan, @ for context, / for commands

Auto

Task4_Sorting_Comparison.py ×

Task4_Sorting_Comparison.py > ...

Review Next File

```python
def builtin_sort(arr):
    return arr_copy


def compare_sorting_algorithms(test_sizes=[100, 500, 1000, 5000]):
    """
    Compare performance of Bubble Sort vs Built-in Sort.

    Args:
        test_sizes (list): List of different array sizes to test
    """
    print("="*70)
    print("PERFORMANCE COMPARISON: Bubble Sort vs Built-in sort()")
    print("="*70)
    print(f"\n{'Array Size':<15} {'Bubble Sort (s)':<20} {'Built-in sort() (s)':<20} {'Speedup':<15}")
    print("-"*70)

    results = []

    for size in test_sizes:
        # Generate random list
        random_list = [random.randint(1, 10000) for _ in range(size)]

        # Test Bubble Sort
        start_time = time.time()
        bubble_result = bubble_sort(random_list)
        bubble_time = time.time() - start_time

        # Test Built-in Sort
        start_time = time.time()
        builtin_result = builtin_sort(random_list)
        builtin_time = time.time() - start_time

        # Calculate speedup
        speedup = bubble_time / builtin_time if builtin_time > 0 else 0

        # Verify both produce same results
        assert bubble_result == builtin_result, "Results don't match!"

        results.append({
            'size': size,
            'bubble_time': bubble_time,
            'builtin_time': builtin_time,
            'speedup': speedup
        })

        print(f"{size:<15} {bubble_time:<20.6f} {builtin_time:<20.6f} {speedup:<15.2f}x")

    print("-"*70)

    return results


def demonstrate_sorting():
    """
    Demonstrate both sorting methods with a small example.
    """
    print("\n" + "="*70)
    print("DEMONSTRATION: Sorting a Sample List")
    print("="*70)

    # Sample unsorted list
    sample_list = [64, 34, 25, 12, 22, 11, 90, 5]
```

AIC
.venv
README_Task2.md
README_Task3.md
README_Task4.md
sample_data.csv
sample_output.txt
Task2_File_Operatio...
Task3_CSV_Data_A...
Task4_Sorting_Com...

Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for sample_data.csv?

Show Recommendations Install

Cursor Tab Ln 243, Col 1 Spaces: 4 UTF-8 LF {} Python 3.9.6 ('.venv': venv)

AIC

New Agent

Agents

Bubble sort vs Pytho... 16m

Bubble sort vs Python's built-in...

16m · +439 · Auto

Task 4: Sorting Lists – Manual vs Built-in
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Plan, @ for context, / for commands

Auto

Task4_Sorting_Comparison.py ×

Task4_Sorting_Comparison.py > ...

Review Next File

```python
    def print_comparison_summary():
    - Extremely fast and efficient
    - Optimized for real-world data patterns
    - Handles various edge cases
    - Well-tested and reliable
  • Disadvantages:
    - Uses more memory than in-place algorithms
    - Less educational value (implementation is hidden)

KEY TAKEAWAY:
-------------
For educational purposes, implementing Bubble Sort helps understand
algorithmic thinking. However, for practical applications, always use
Python's built-in sort() as it's significantly faster and more reliable.

Performance Difference:
• Built-in sort() is typically 100-1000x faster than Bubble Sort
• The difference becomes exponentially larger as dataset size increases
"""

    print(comparison)


def main():
    """
    Main function to run the sorting comparison.
    """
    print("\n" + "="*70)
    print("Task 4: Sorting Lists – Manual vs Built-in")
    print("Generated using AI Assistant")
    print("="*70)

    # Demonstrate with small example
    demonstrate_sorting()

    # Performance comparison
    print("\n")
    compare_sorting_algorithms()

    # Print comparison summary
    print_comparison_summary()

    print("\n" + "="*70)
    print("✅ Task 4 completed successfully!")
    print("="*70 + "\n")


if __name__ == "__main__":
    main()
    Kl to chat, ⌘K to generate
```

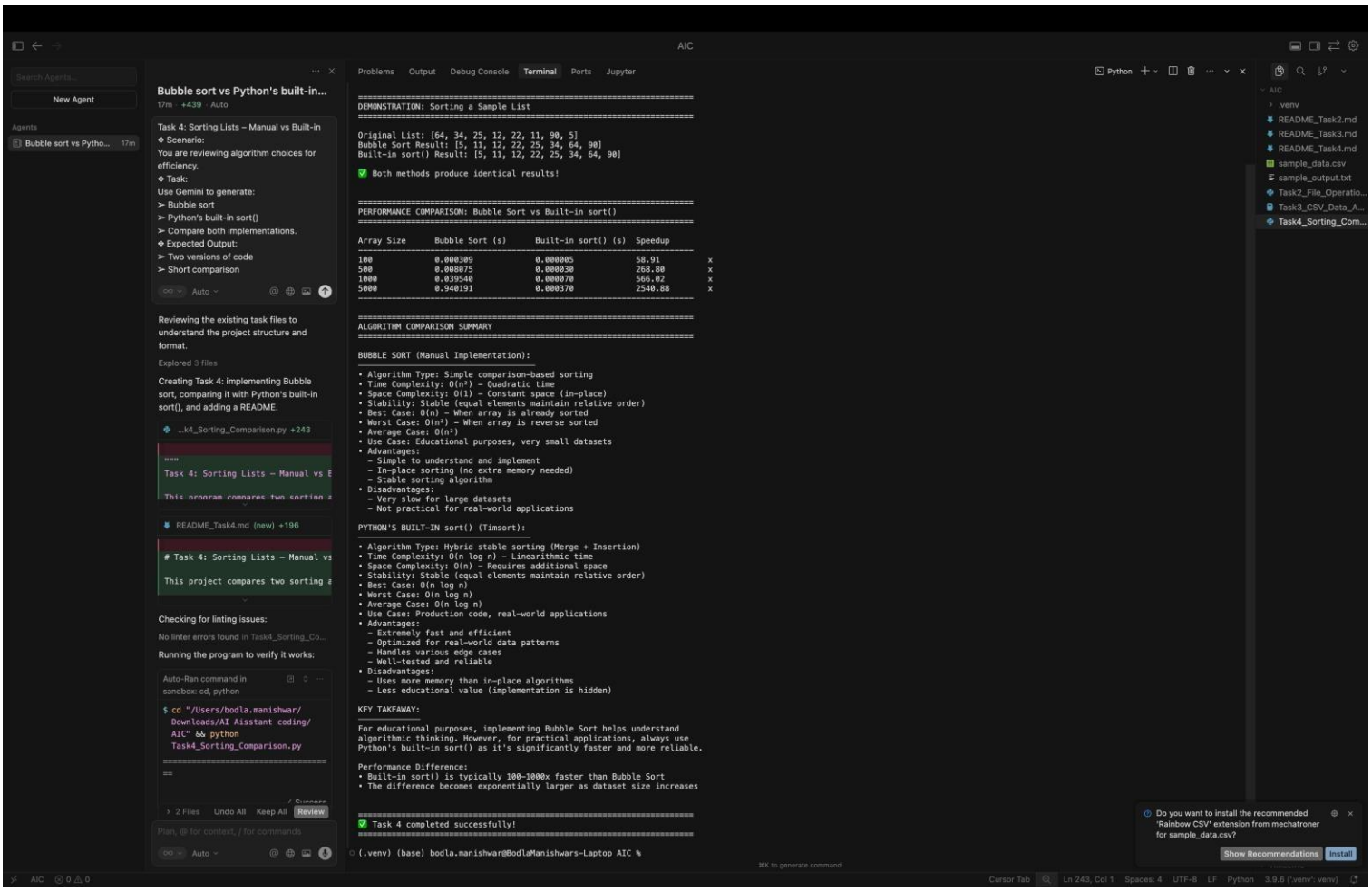
AIC
.venv
README_Task2.md
README_Task3.md
README_Task4.md
sample_data.csv
sample_output.txt
Task2_File_Operatio...
Task3_CSV_Data_A...
Task4_Sorting_Com...

Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for sample_data.csv?

Show Recommendations Install

Cursor Tab Ln 243, Col 1 Spaces: 4 UTF-8 LF {} Python 3.9.6 ('.venv': venv)

OUTPUT:


AIC
Bubble sort vs Python's built-in...
Task 4: Sorting Lists – Manual vs Built-in
Scenario:
You are reviewing algorithm choices for efficiency.
Task:
Use Gemini to generate:
Bubble sort
Python's built-in sort()
Compare both implementations.
Expected Output:
Two versions of code
Short comparison
Reviewing the existing task files to understand the project structure and format.
Creating Task 4: implementing Bubble sort, comparing it with Python's built-in sort(), and adding a README.
Checking for linting issues:
Running the program to verify it works:
Problems Output Debug Console Terminal Ports Jupyter
DEMONSTRATION: Sorting a Sample List
Original List: [64, 34, 25, 12, 22, 11, 90, 5]
Bubble Sort Result: [5, 11, 12, 22, 25, 34, 64, 90]
Built-in sort() Result: [5, 11, 12, 22, 25, 34, 64, 90]
Both methods produce identical results!
PERFORMANCE COMPARISON: Bubble Sort vs Built-in sort()
Array Size Bubble Sort (s) Built-in sort() (s) Speedup
100 0.000309 0.000005 58.91 x
500 0.008075 0.000030 268.80 x
1000 0.039540 0.000070 566.02 x
5000 0.940191 0.000370 2540.88 x
ALGORITHM COMPARISON SUMMARY
BUBBLE SORT (Manual Implementation):
• Algorithm Type: Simple comparison-based sorting
• Time Complexity: O(n²) – Quadratic time
• Space Complexity: O(1) – Constant space (in-place)
• Stability: Stable (equal elements maintain relative order)
• Best Case: O(n) – When array is already sorted
• Worst Case: O(n²) – When array is reverse sorted
• Average Case: O(n²)
• Use Case: Educational purposes, very small datasets
• Advantages:
– Simple to understand and implement
– In-place sorting (no extra memory needed)
– Stable sorting algorithm
• Disadvantages:
– Very slow for large datasets
– Not practical for real-world applications
PYTHON'S BUILT-IN sort() (Timsort):
• Algorithm Type: Hybrid stable sorting (Merge + Insertion)
• Time Complexity: O(n log n) – Linearithmic time
• Space Complexity: O(n) – Requires additional space
• Stability: Stable (equal elements maintain relative order)
• Best Case: O(n log n)
• Worst Case: O(n log n)
• Average Case: O(n log n)
• Use Case: Production code, real-world applications
• Advantages:
– Extremely fast and efficient
– Optimized for real-world data patterns
– Handles various edge cases
– Well-tested and reliable
• Disadvantages:
– Uses more memory than in-place algorithms
– Less educational value (implementation is hidden)
KEY TAKEAWAY:
For educational purposes, implementing Bubble Sort helps understand algorithmic thinking. However, for practical applications, always use Python's built-in sort() as it's significantly faster and more reliable.
Performance Difference:
• Built-in sort() is typically 100-1000x faster than Bubble Sort
• The difference becomes exponentially larger as dataset size increases
Task 4 completed successfully!
(.venv) (base) bodla.manishwar@BodlaManishwars-Laptop AIC %
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CODE EXPLANATION:

This program compares Bubble Sort and Python's built-in sort(). Bubble Sort manually compares and swaps elements to arrange them in order, but it is slow for large lists because it has $O(n^2)$ time complexity. Python's built-in sort() uses an efficient algorithm and sorts data much faster with O(n log n) time complexity. The program measures execution time for both methods and shows that the built-in sort is much faster and more suitable for real-world use.